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3.2.S.4.1 Specification

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Sitagliptin Phosphate (STP No.: STP/FP/SP/002-08) final API release specification appended in the following pages.

 Lee Pharma Limited Unit-I, Gaddapotharam	SPECIFICATIONS AND TEST PROCEDURE	
	STP No. : STP/FP/SP/002-08	Effective Date:
	Supersedes : SSPFP02 Rev. No. 07	10 / 07 / 2024
	Department : QUALITY CONTROL	Page No. : 1 of 41
Product Name : Sitagliptin Phosphate		
Molecular Formula : C ₁₆ H ₁₅ F ₆ N ₅ O.H ₃ PO ₄ . H ₂ O		Product Code : SP
Molecular Weight : 523.33		Reference : USP / In - House

A. SPECIFICATIONS:

S.No.	Test Parameters	Specifications	Reference to test method
1.0	Description	White or almost white powder.	< In - House >
2.0	Solubility	Soluble in water, very slightly soluble in anhydrous ethanol, practically insoluble in heptane.	< In - House >
3.0	Identification by:		
	a) IR	The Infra-red spectrum of sample should be concordant to that of Sitagliptin Phosphate CRS / Working standard.	USP < 197K >
	b) Enantiomeric purity by HPLC	Compare the retention time of major peak obtained in sample chromatogram should be match with that of chromatogram of system suitability solution.	< USP method >
	c) Phosphate test	Meets the requirements of test A of Orthophosphates.	USP < 191 >
4.0	Water content by KF (% , w/w)	Between 3.3 and 3.7	USP < 921> Method Ia
5.0	Residue on ignition (% , w/w)	Not more than 0.20	< In - House >

	Prepared by	Reviewed by	Approved by
Name	A. Kalpana	R. Ramalingeswara Rao	K. Siva Santhosh Kumar
Designation & Department	Asst. Manager - QC	Dy. Manager - QC	Asst. Manager - QA
Signature & Date	 10/07/2024	 10/07/2024	 10/07/24

 Lee Pharma Limited Unit-I, Gaddapotharam	SPECIFICATIONS AND TEST PROCEDURE		
	STP No. : STP/FP/SP/002-08		Effective Date: 10 / 07 / 2024
	Supersedes : SSPFP02 Rev. No. 07		
	Department : QUALITY CONTROL		Page No. : 2 of 41
Product Name : Sitagliptin Phosphate			
Molecular Formula : C ₁₆ H ₁₅ F ₆ N ₅ O.H ₃ PO ₄ . H ₂ O		Product Code : SP	
Molecular Weight : 523.33		Reference : USP / In - House	
S.No.	Test Parameters	Specifications	Reference to test method
6.0	Phosphate content by potentiometry (On anhydrous basis) (%)	Between 18.5 and 20.5	< In - House validated method >
7.0	Enantiomeric purity by HPLC (%) a) S - Enantiomer	Not more than 0.50	< USP method >
8.0	Organic impurities by HPLC (%) a) Any individual impurity b) Total impurities	Not more than 0.10 Not more than 0.50	< USP method >
9.0	Assay by HPLC (On anhydrous and solvent free basis) (% , w/w)	Not less than 98.0 and Not more than 102.0	< USP method >
10.0	Residual solvents by GC (ppm) a) Methanol b) Acetone c) Isopropyl alcohol d) Acetonitrile e) Methylene dichloride f) Methyl tertiary butyl ether g) Toluene h) n-Butanol i) Chloroform	Not more than 3000 Not more than 5000 Not more than 5000 Not more than 410 Not more than 600 Not more than 5000 Not more than 890 Not more than 5000 Not more than 60	< In - House validated method >
11.0	Acetic acid content by HPLC (ppm)	Not more than 5000	< In - House validated method >
12.0	NTPy / NTTP Nitrosamine impurity content by LC - MS (ppm)	Not more than 0.29	< In - House validated method >

	Prepared by	Reviewed by	Approved by
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Designation & Department	Asst. Manager - QC	Dy. Manager - QC	Asst. Manager - QA
Signature & Date	 10/07/2024	 10/07/24	 10/07/24

 Lee Pharma Limited Unit-I, Gaddapotharam	SPECIFICATIONS AND TEST PROCEDURE	
	STP No. : STP/FP/SP/002-08	Effective Date:
	Supersedes : SSPFP02 Rev. No. 07	10 / 07 / 2024
	Department : QUALITY CONTROL	Page No. : 3 of 41
Product Name : Sitagliptin Phosphate		
Molecular Formula : C ₁₆ H ₁₅ F ₆ N ₅ O.H ₃ PO ₄ . H ₂ O		Product Code : SP
Molecular Weight : 523.33		Reference : USP / In - House
Storage: Store in well - closed containers at below 25°C, excursion is permitted between 15°C to 30°C.		
Packaging: The material shall be packed in double lined low - density food grade polyethylene bags (transparent inner and black outer) tie with plastic fastener and followed by HDPE drums.		

Product / Impurity name	IUPAC name
Sitagliptin Phosphate	(3R)-3-amino-1-[3-(trifluoromethyl)-5,6-dihydro[1,2,4]triazolo [4,3-a]pyrazin-7(8H)-yl]-4-(2,4,5-trifluorophenyl)butan-1-one phosphate monohydrate.
NTTP/ NTPy/ Nitroso-STG-19	7-nitroso-3-(trifluoro methyl)-5,6,7,8-tetra hydro-[1,2,4] triazolo [4,3-a] pyrazine

	Prepared by	Reviewed by	Approved by
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Signature & Date	 10/07/2024	 10/07/24	 10/07/24

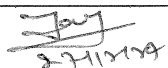
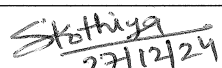
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Effective Date: 25.03.2024

**NEXT WAVE (INDIA)**RAMPUR GHAT ROAD, PAONTA SAHIB
SIRMOUR (HP) – 173025**MASTER****RAW MATERIAL STANDARD TESTING SPECIFICATION**

MATERIAL NAME		DEPARTMENT	
Sitagliptin Phosphate USP		RESEARCH & DEVELOPMENT	
STS No.	RM/STS/NFRD/USP/003	Revision No.	01
STP No.	RM/STP/NFRD/USP/003	Supersedes	00
Effective Date	30/12/2024	Page No.	Page 1 of 2

Sr. No.	TEST	SPECIFICATION
1.0	Description	White or almost white powder.
2.0	Solubility	Soluble in water, very slightly soluble in anhydrous ethanol and practically insoluble in heptane.
3.0	Identification	
	By IR	The Infrared absorption spectrum of the sample Should be concordant to that of Sitagliptin phosphate standard.
	Enantiomeric Purity By HPLC	The principal peak in the chromatogram obtained with the sample solution should be correspond to the principal peak in the chromatogram obtained with system Suitability solution.
	Phosphate Test	Meets the requirements of test A of Orthophosphates.
4.0	Assay by HPLC (On anhydrous and Solvent-free basis)	98.0% to 102.0%
5.0	Organic Impurities by HPLC	
	Any individual impurity	NMT 0.1%
	Total impurities	NMT 0.5%
6.0	Enantiomeric Purity by HPLC	
	S-enantiomer	NMT 0.5%
7.0	Water content (by KF)	NMT 3.3% to 3.7% w/w
8.0	*Residue on Ignition	NMT 0.20% w/w
9.0	*Residual Solvents	

	PREPARED BY	CHECKED BY	APPROVED BY
SIGNATURE/ DATE	 27/12/24	 27/12/24	 28/12/24
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DESIGNATION	Sr. Officer	Executive	Dy. Manager QA

SOP No: R&D-026

Format No- R&DF/026/02

UNCONTROLLED



NEXT WAVE (INDIA)

RAMPUR GHAT ROAD, PAONTA SAHIB
SIRMOUR (HP) – 173025

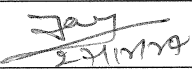
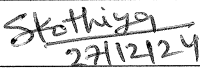
MASTER

RAW MATERIAL STANDARD TESTING SPECIFICATION

MATERIAL NAME		DEPARTMENT	
Sitagliptin Phosphate USP		RESEARCH & DEVELOPMENT	
STS No.	RM/STS/NFRD/USP/003	Revision No.	01
STP No.	RM/STP/NFRD/USP/003	Supersedes	00
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	Methanol	NMT 3000 ppm
	Acetone	NMT 5000 ppm
	Isopropyl Alcohol	NMT 5000 ppm
	Acetonitrile	NMT 410 ppm
	Methylene dichloride	NMT 600 ppm
	Methyl tertiary butyl ether	NMT 5000 ppm
	Toluene	NMT 890 ppm
	n-Butanol	NMT 5000 ppm
	Chloroform	NMT 60 ppm
10.0	*Acetic acid content by HPLC	NMT 5000 ppm
11.0	*Phosphate content by Potentiometry (On anhydrous basis)	18.5% to 20.5%
12.0	*NTPy / NTTP Nitrosamine impurity content by LC-MS	NMT 0.29 ppm

*Additional Test

	PREPARED BY	CHECKED BY	APPROVED BY
SIGNATURE/ DATE	 23/12/24	 23/12/24	 23/12/24
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SOP No: R&D-026

Format No- R&DF/026/02

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